EDS Theory :
Assignment No.2

Name : Pranil Pramod Sangale

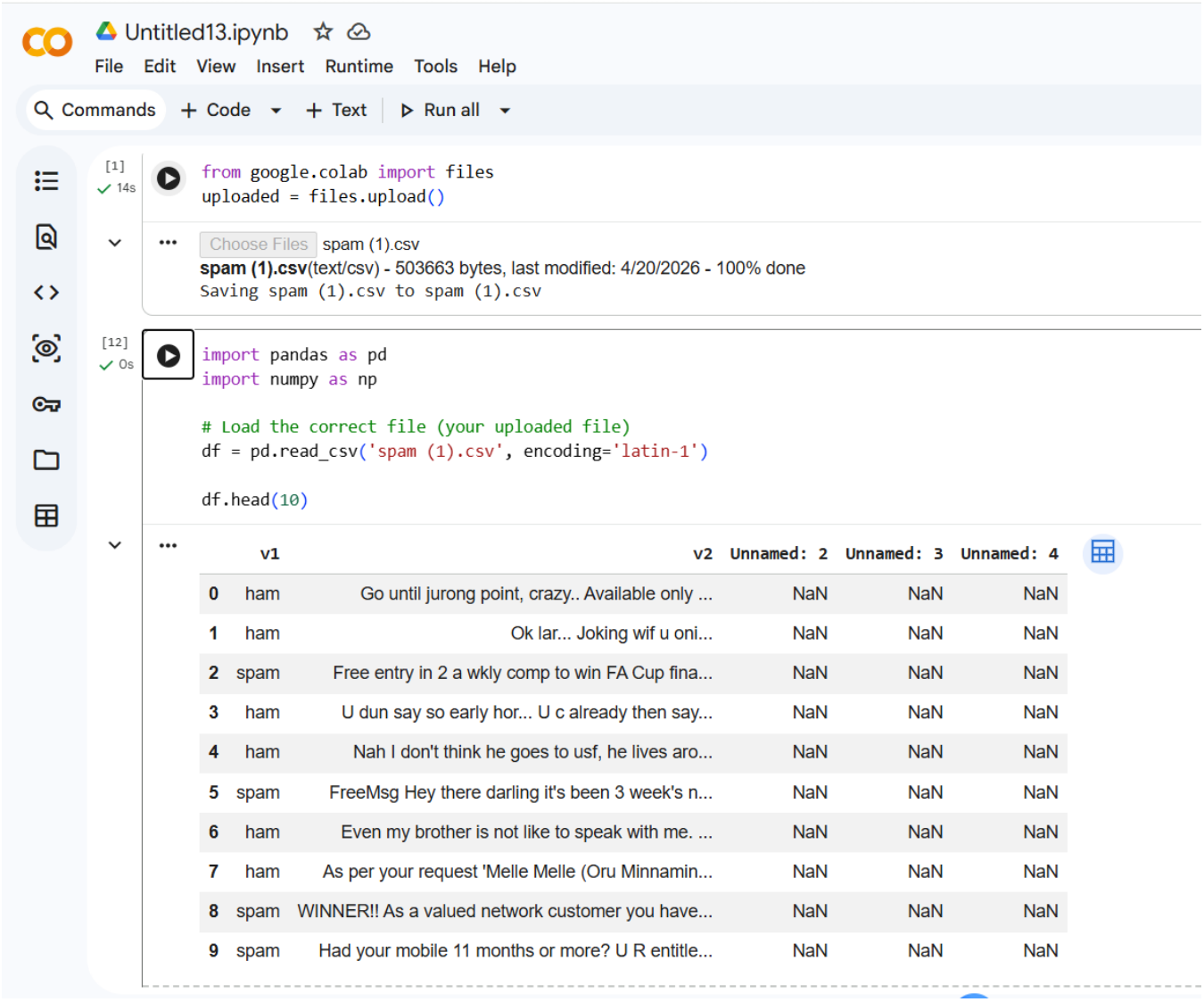
PRN : 202501080077

Subject : EDS

Formulate 20 problem statements for a given dataset using Numpy and Pandas and Apply Numpy and pandas methods to find the solution for the formulated problem statements.

Dataset : SMS Spam Collection

Step 1: Import Libraries and load Dataset

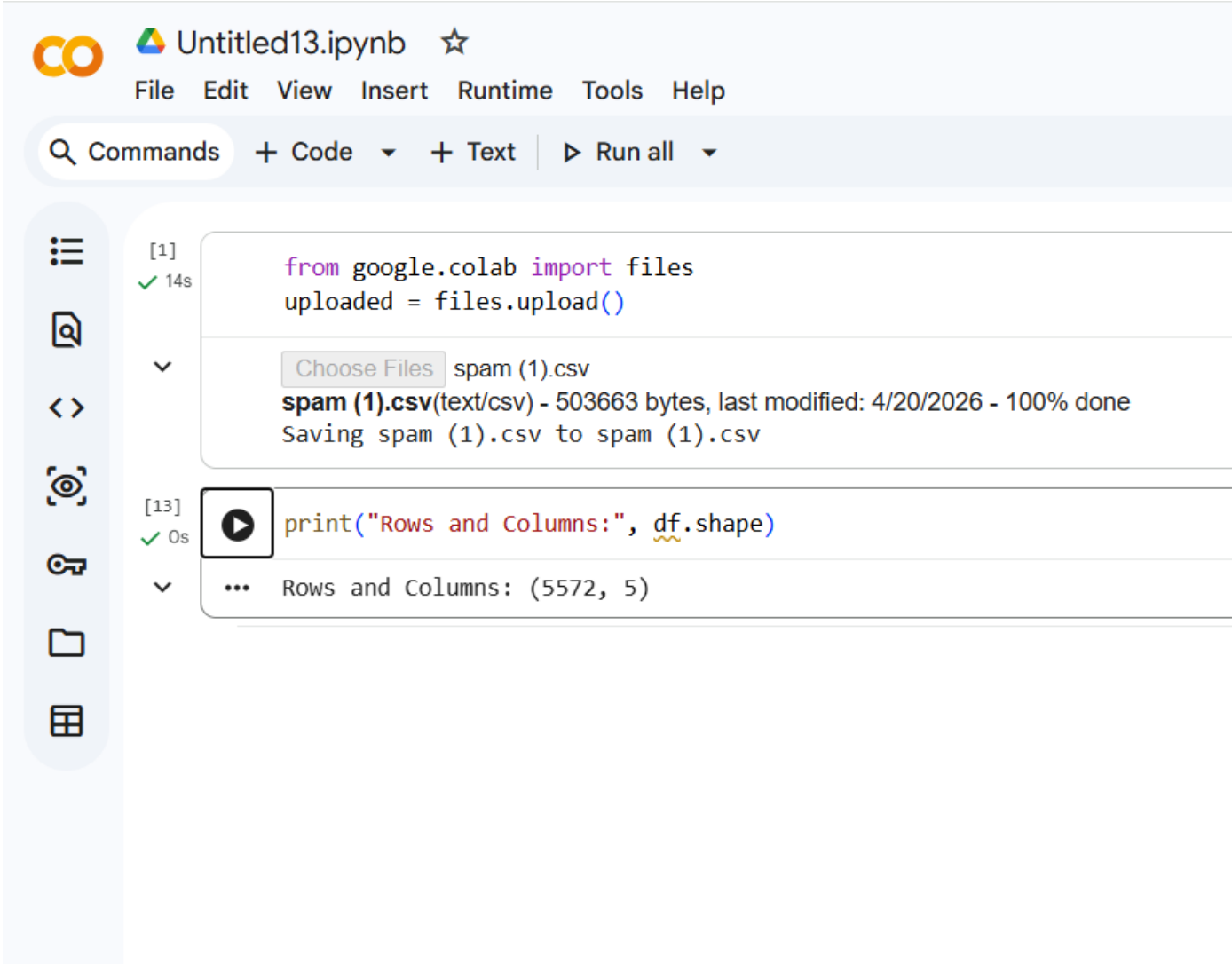


	v1	v2	Unnamed: 2	Unnamed: 3	Unnamed: 4
0	ham	Go until jurong point, crazy.. Available only ...	NaN	NaN	NaN
1	ham	Ok lar... Joking wif u oni...	NaN	NaN	NaN
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	NaN	NaN	NaN
3	ham	U dun say so early hor... U c already then say...	NaN	NaN	NaN
4	ham	Nah I don't think he goes to usf, he lives aro...	NaN	NaN	NaN
5	spam	FreeMsg Hey there darling it's been 3 week's n...	NaN	NaN	NaN
6	ham	Even my brother is not like to speak with me. ...	NaN	NaN	NaN
7	ham	As per your request 'Melle Melle (Oru Minnamin...	NaN	NaN	NaN
8	spam	WINNER!! As a valued network customer you have...	NaN	NaN	NaN
9	spam	Had your mobile 11 months or more? U R entitle...	NaN	NaN	NaN

Step 2 : 20+ Problem Statements and Solutions

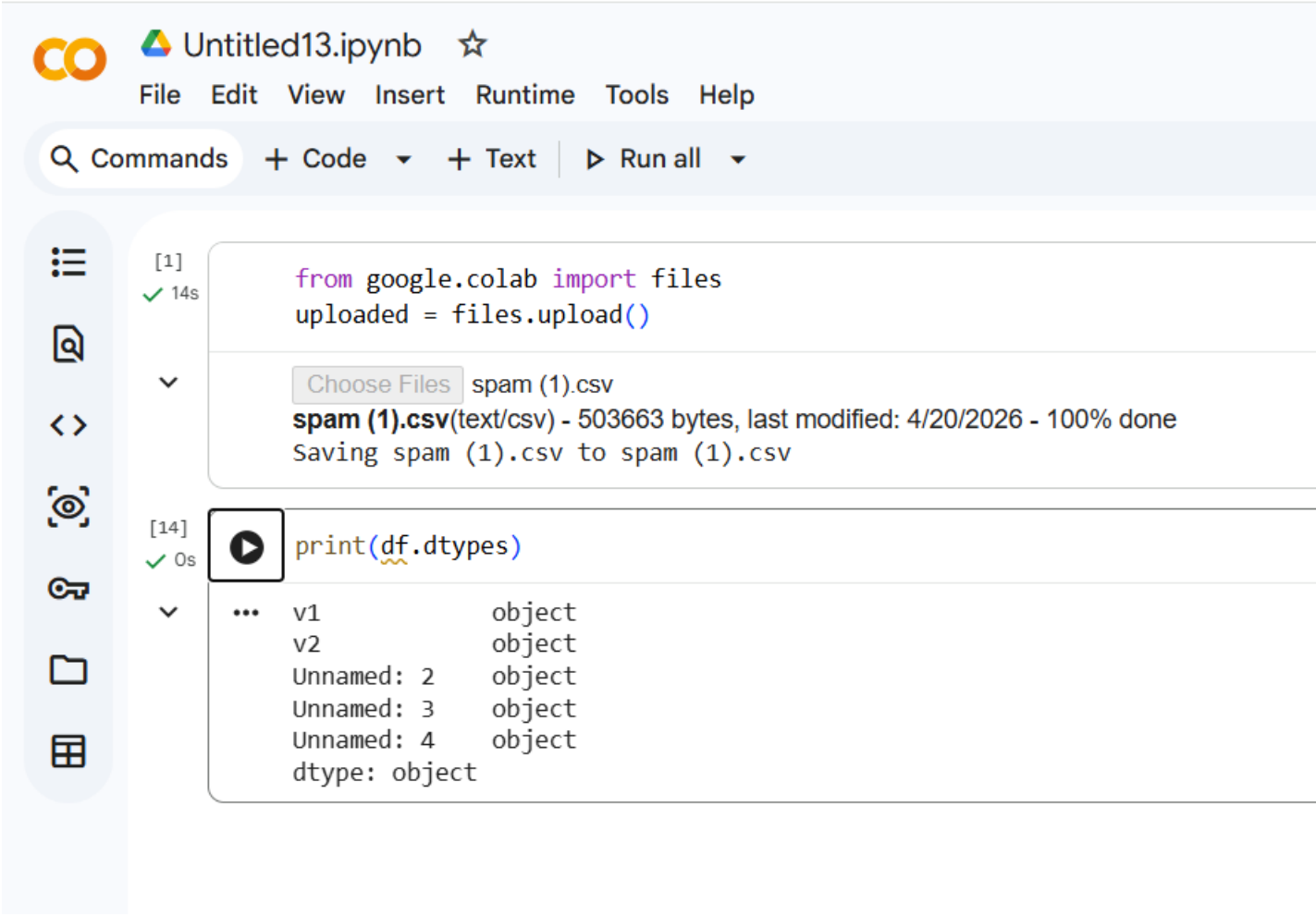
1. Shape of dataset

Problem Statement: Find the total number of messages (rows) and features (columns) in the dataset.



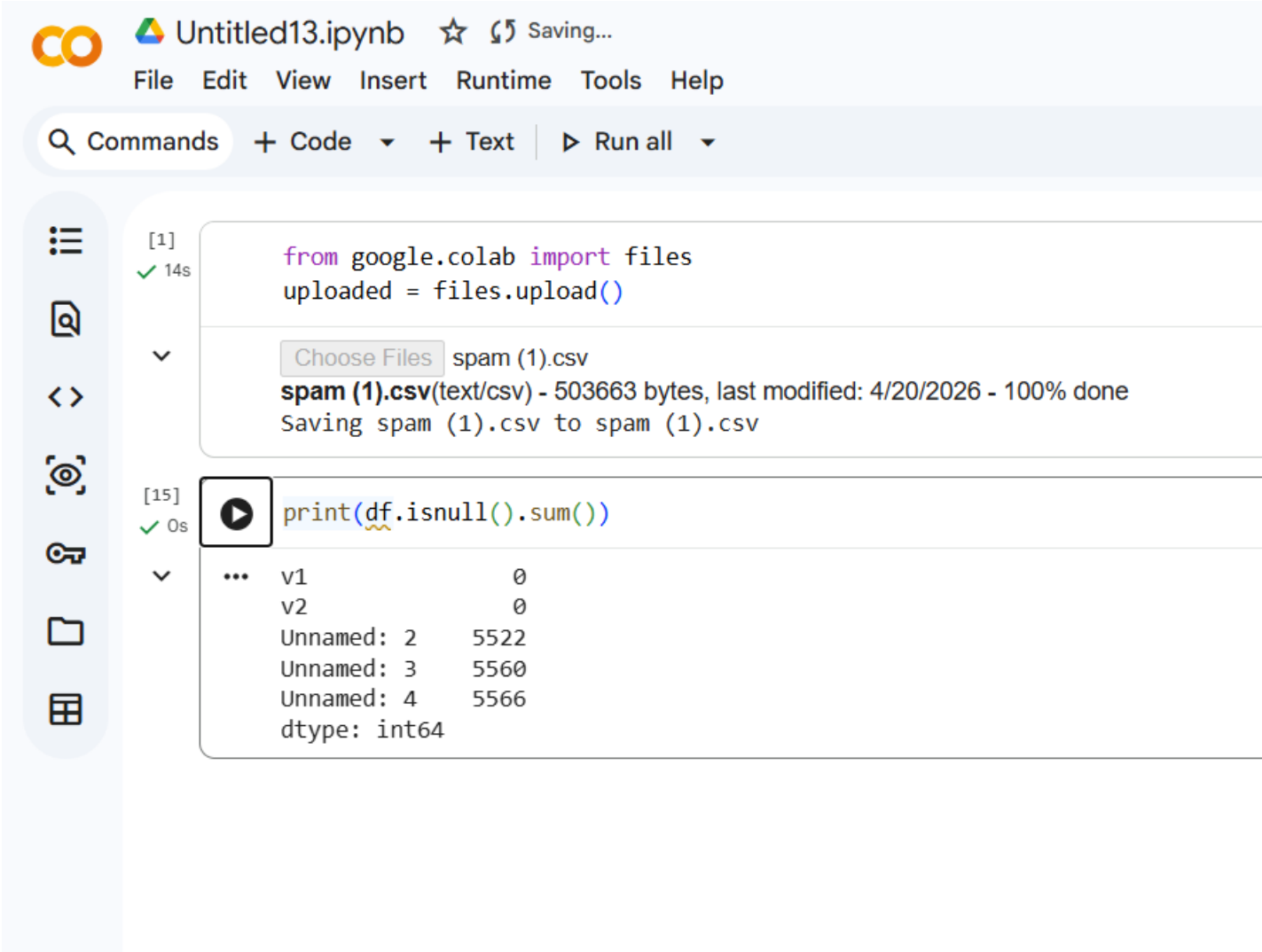
2. DATA TYPES

PROBLEM STATEMENT: Display the column names and their data types.



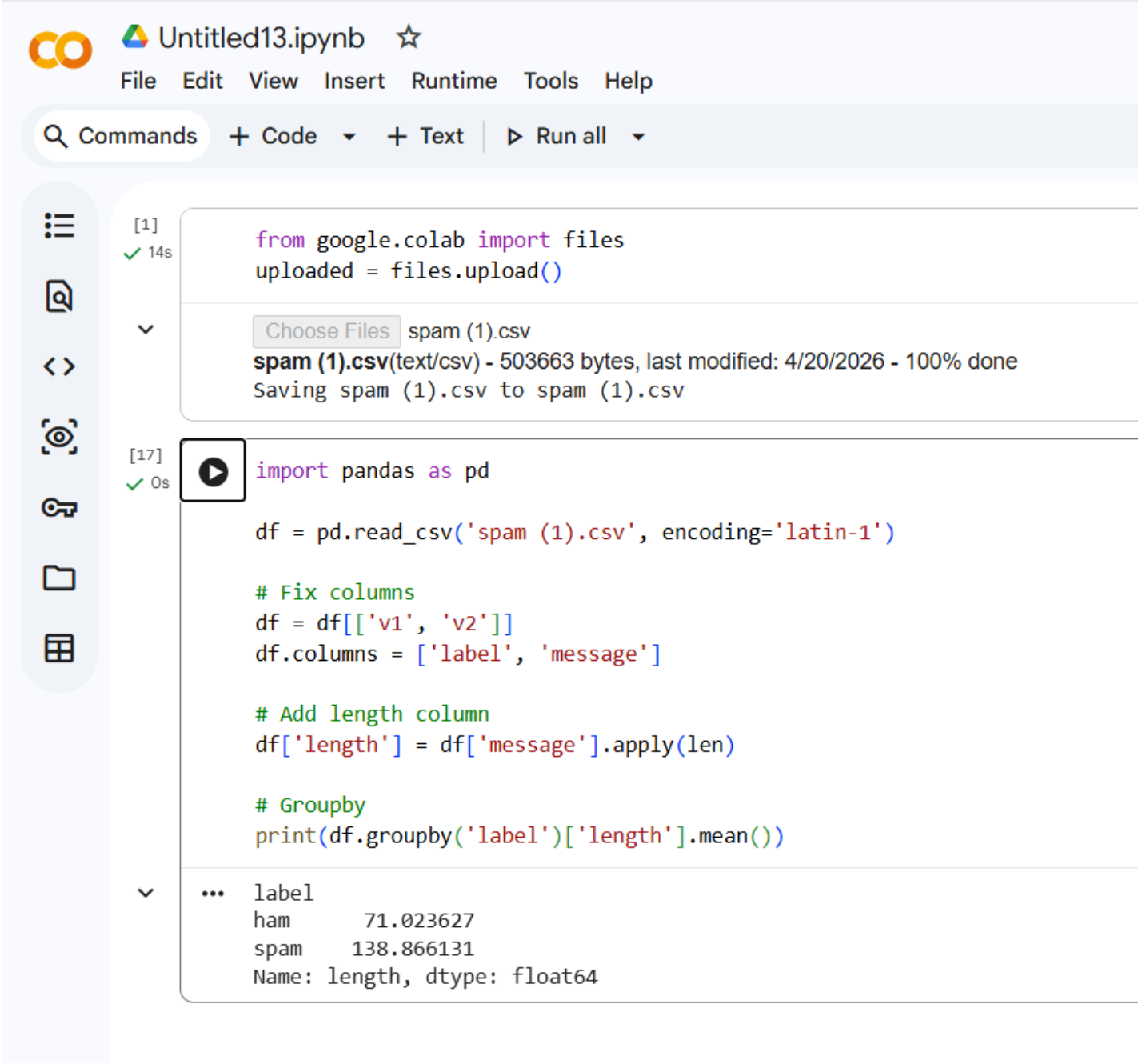
3 MISSING VALUES

PROBLEM STATEMENT: Check how many missing values are present in each column.



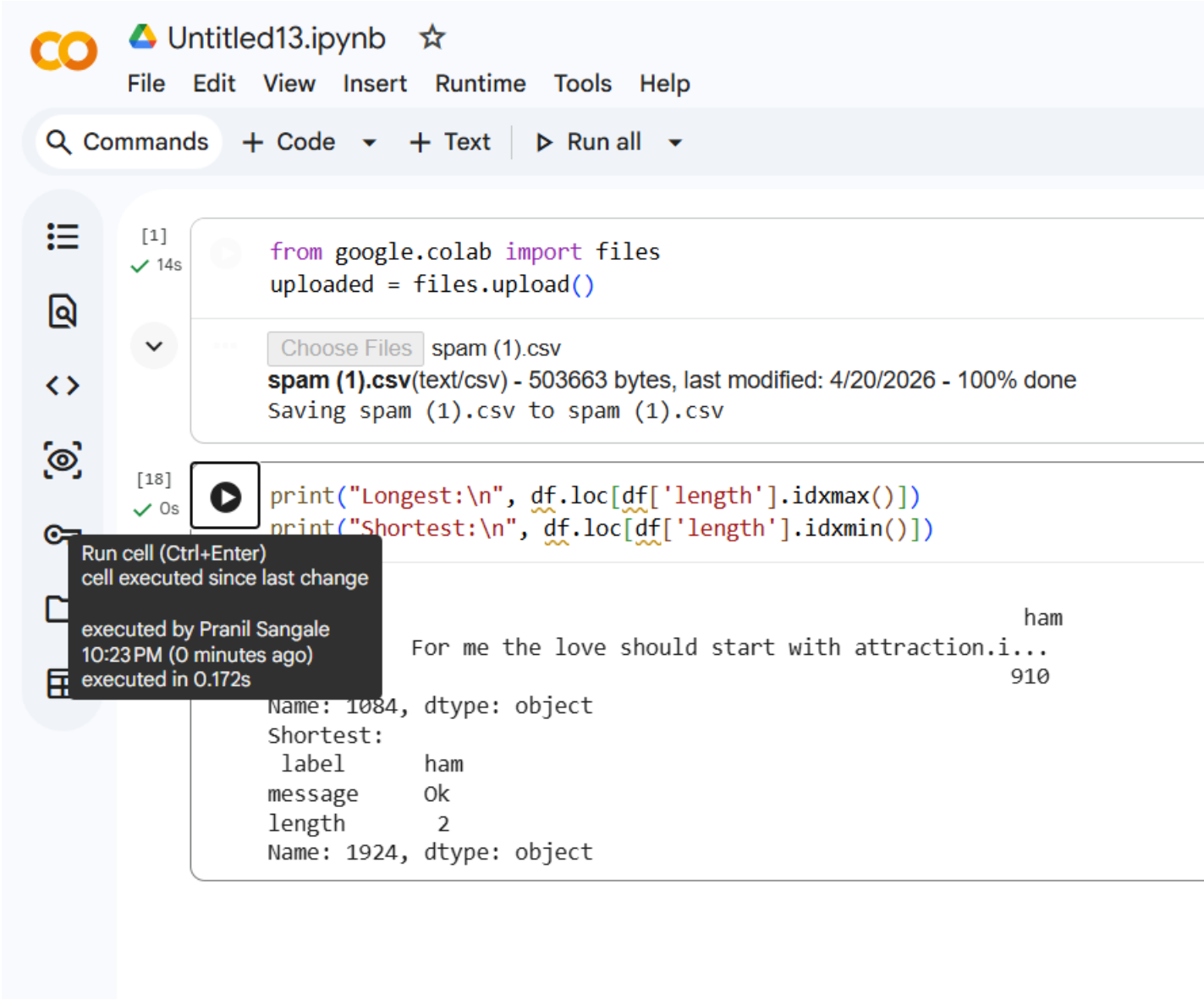
4. GROUP STATISTICS

PROBLEM STATEMENT: Group the dataset by label and calculate count, mean message length.



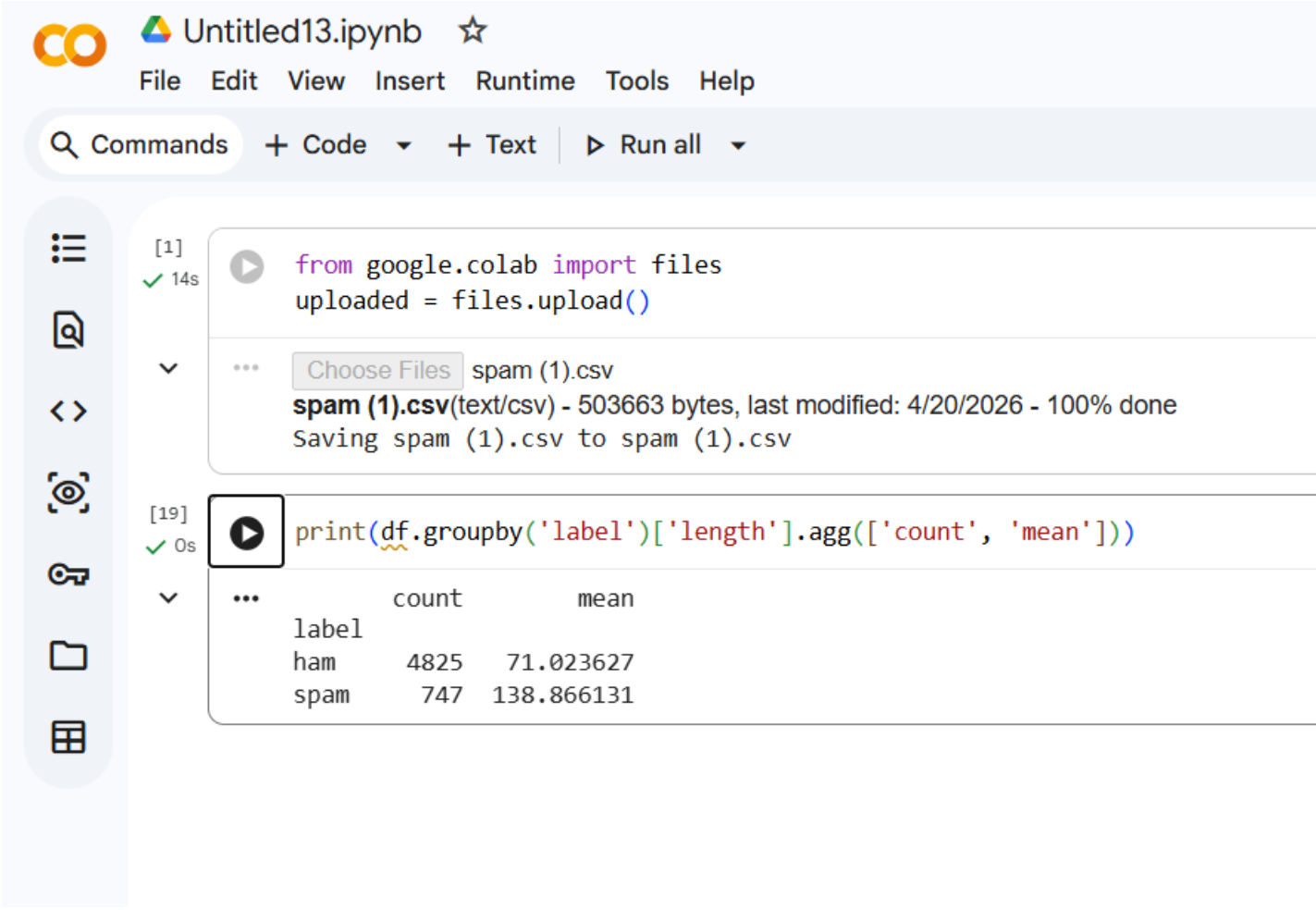
5. LONGEST AND SHORTEST MESSAGE

PROBLEM STATEMENT: Determine the longest and shortest message in the dataset.



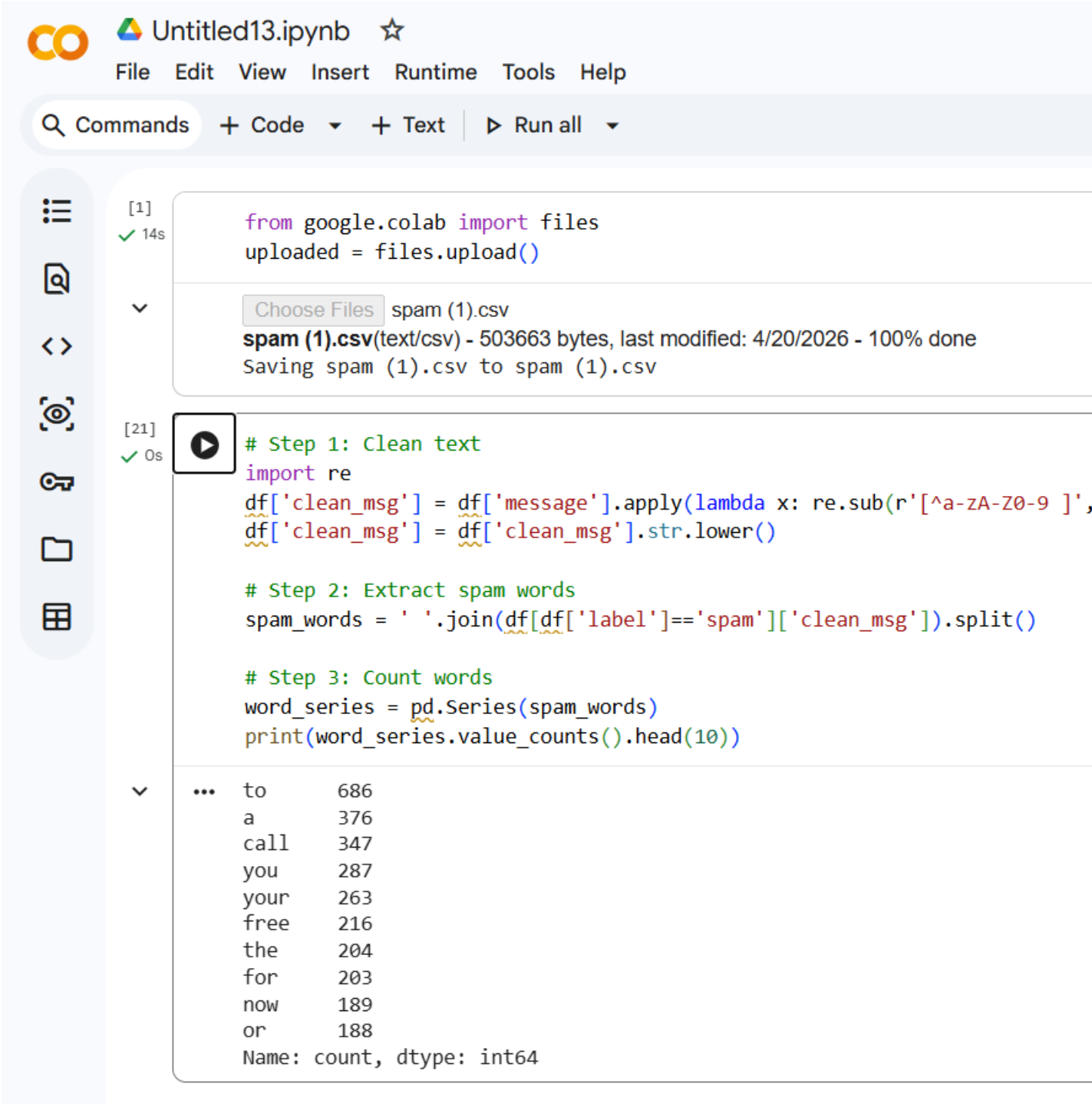
6. AVERAGE MESSAGE LENGTH

PROBLEM STATEMENT: Find the average message length for spam and ham messages separately.



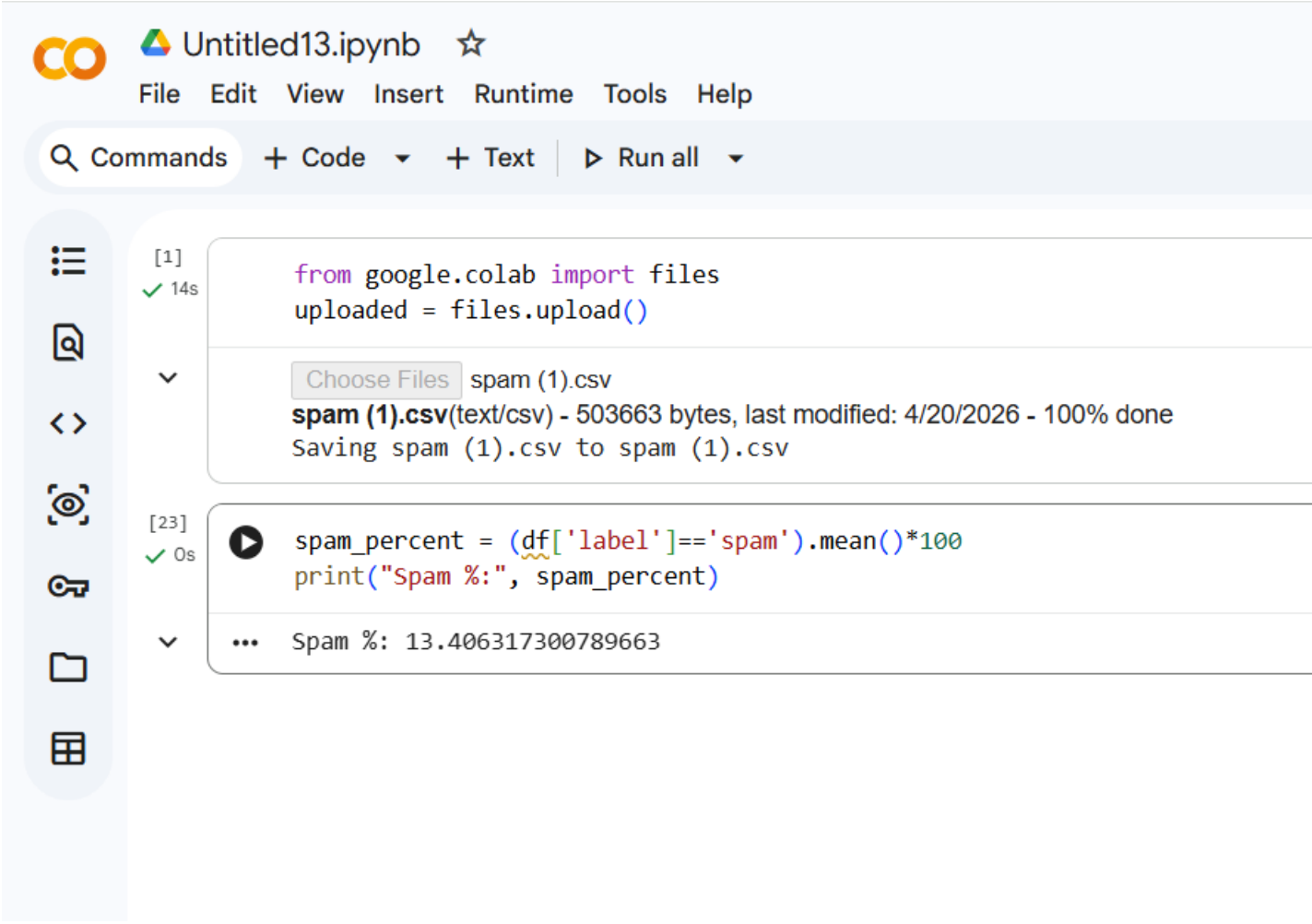
7. Top 10 frequent words in spam

PROBLEM STATEMENT: Find the top 10 most frequent words in spam messages.



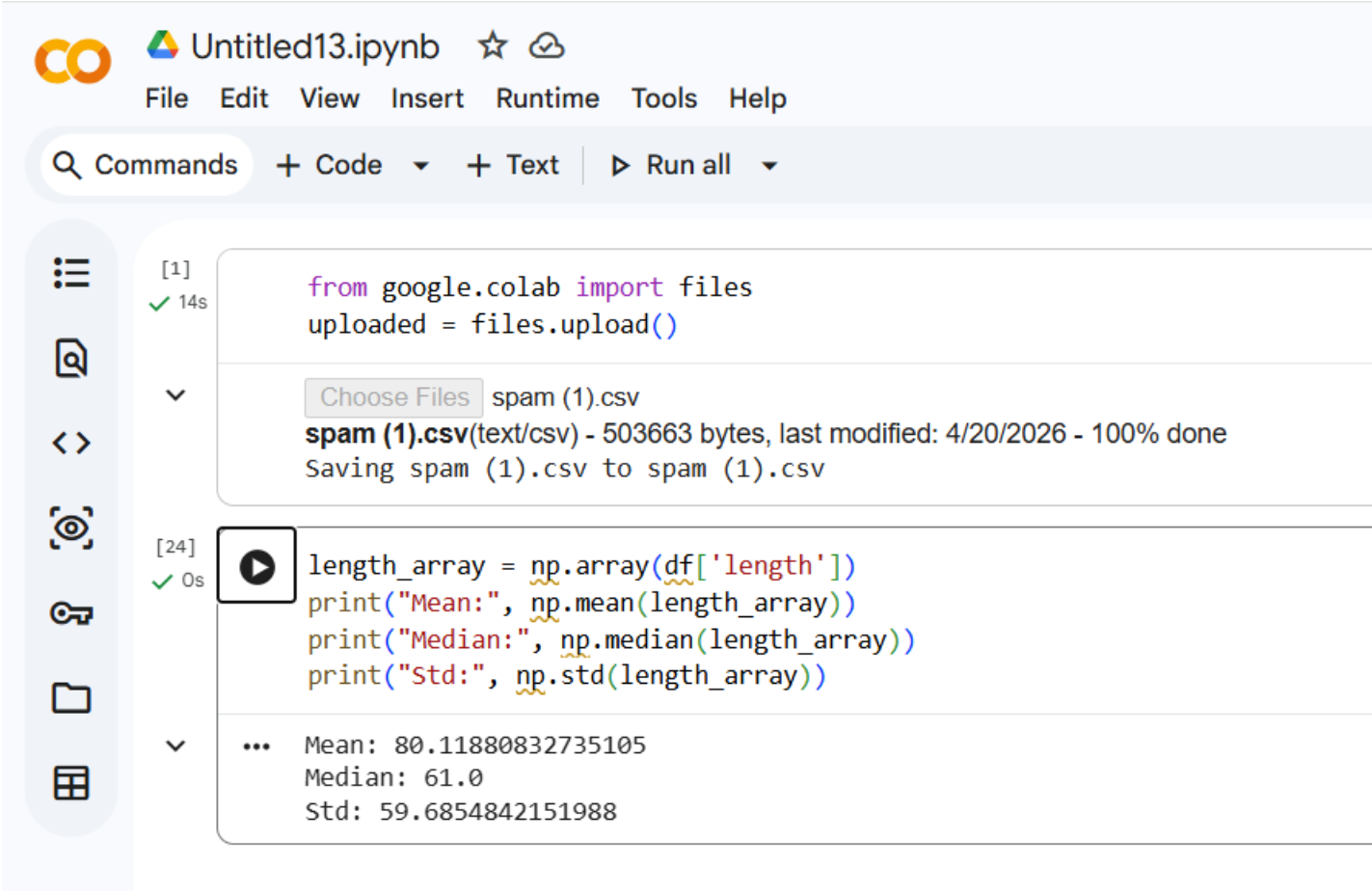
8. Percentage of spam

PROBLEM STATEMENT: Find the percentage of spam messages in the dataset.



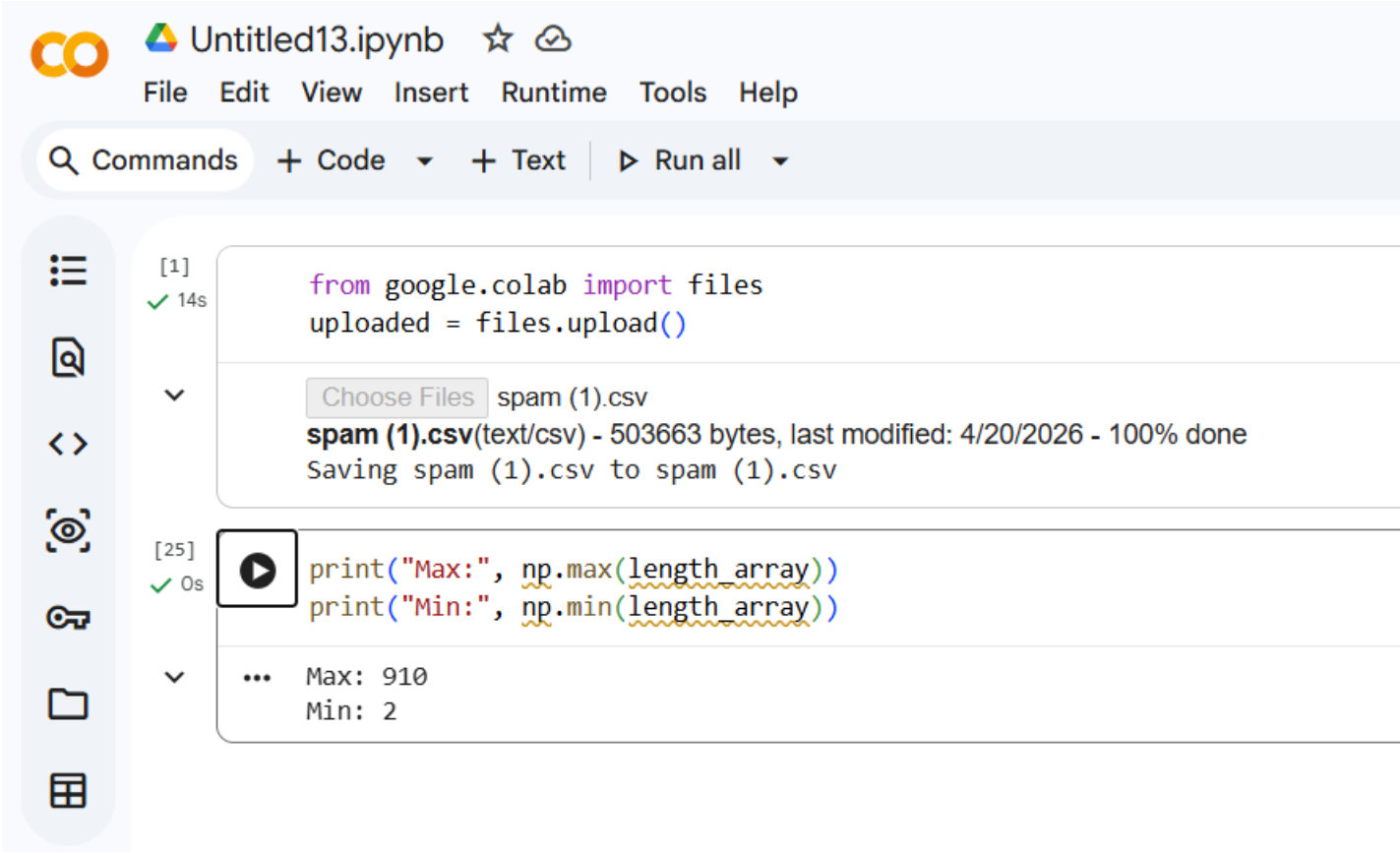
9. MEAN,MEDIAN,STD

PROBLEM STATEMENT: Convert the message length column into a NumPy array and compute mean, median, and standard deviation.



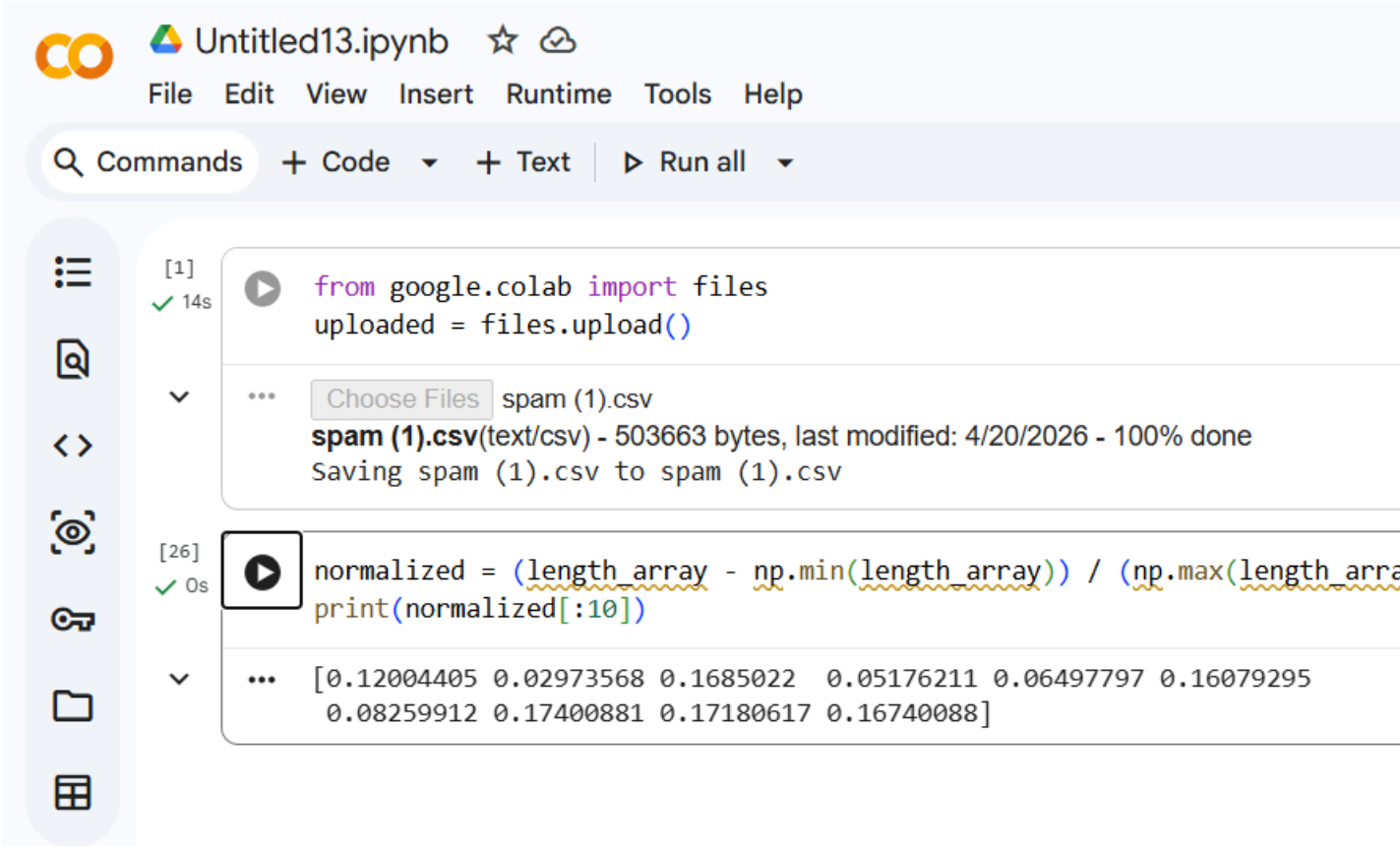
10. MAX , MIN

PROBLEM STATEMENT: Find the maximum and minimum message length using NumPy.



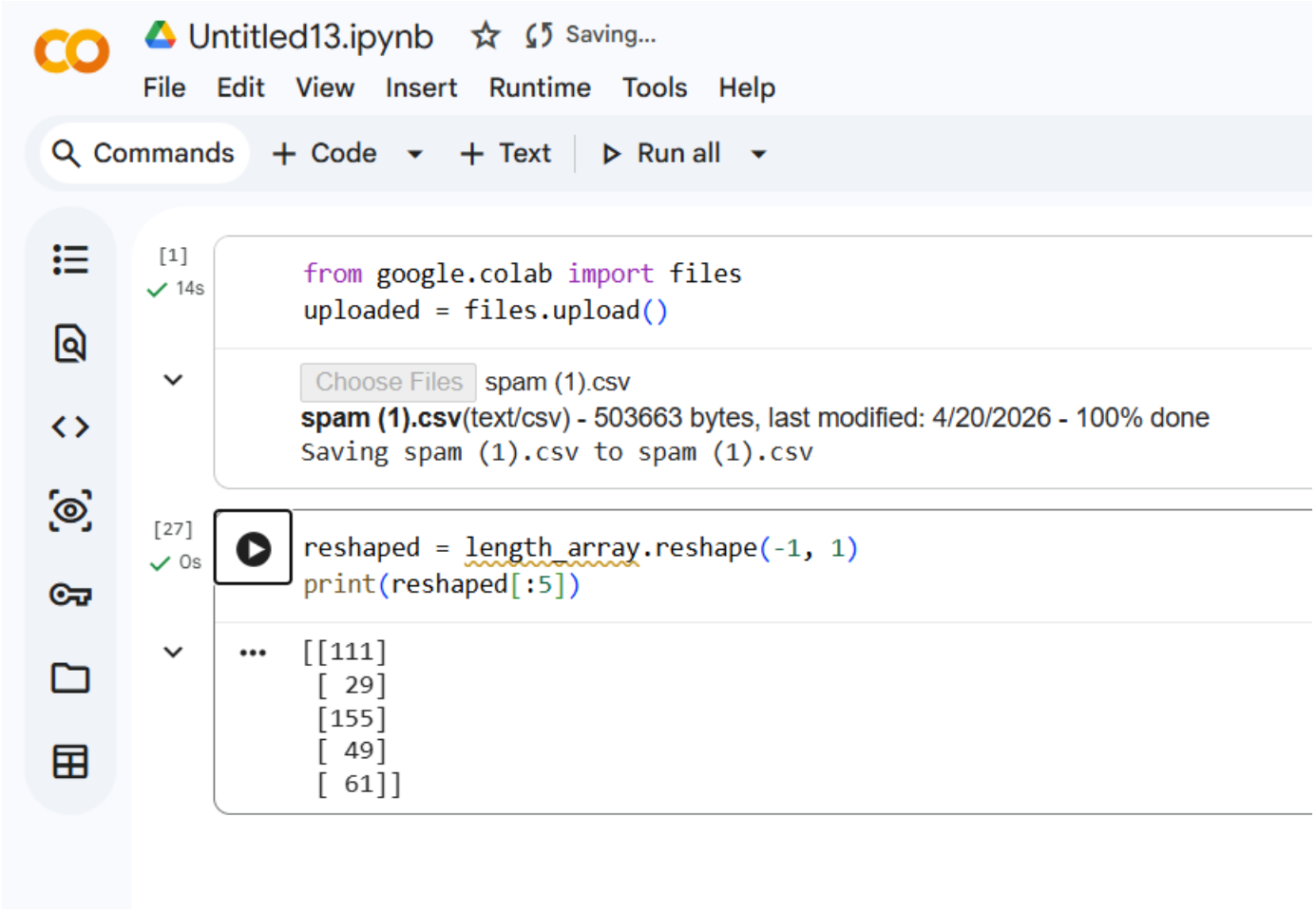
11. NORMALIZE LENGTH

PROBLEM STATEMENT: Normalize the message length using NumPy operations.



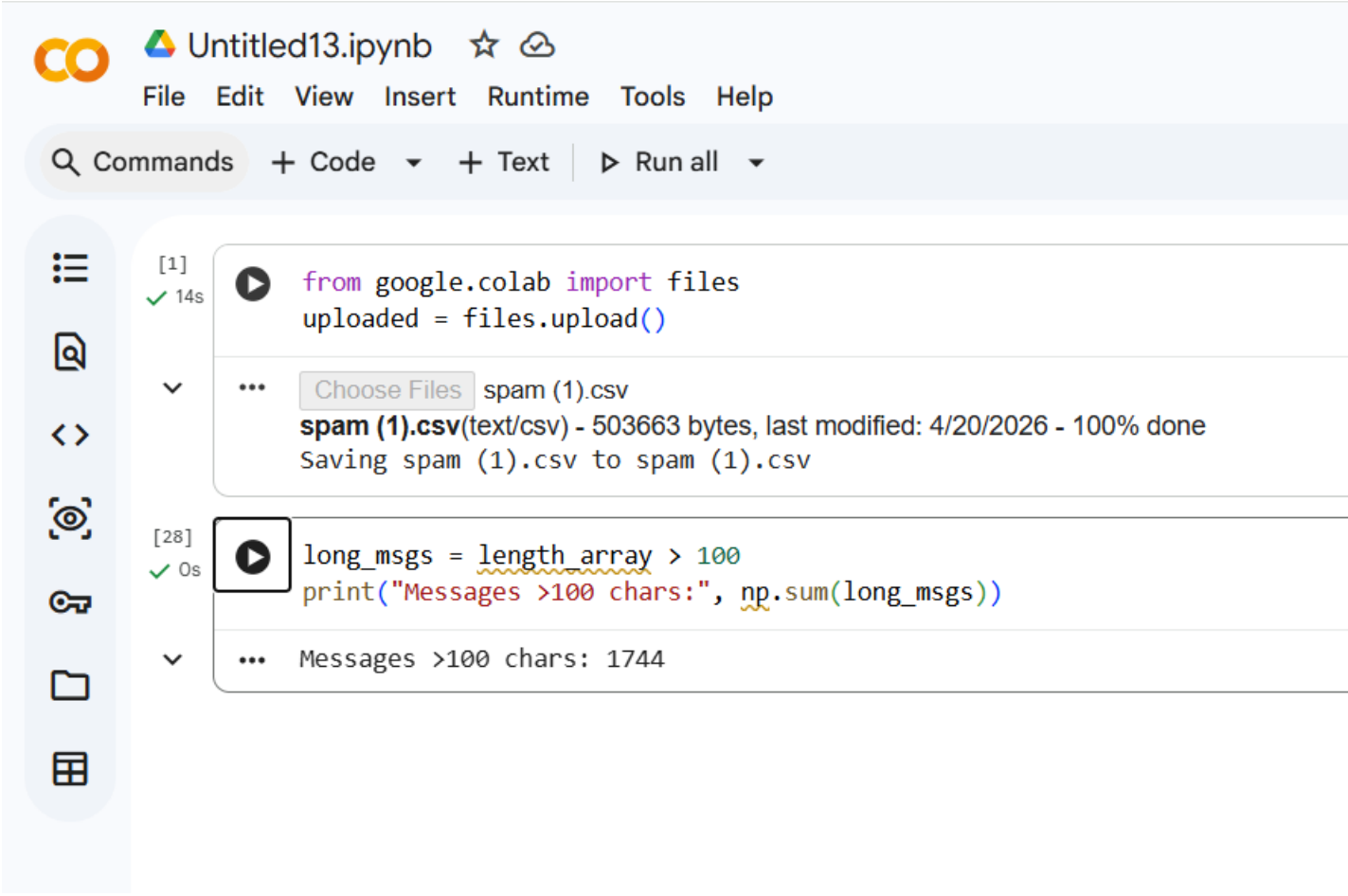
12. RESHAPE ARRAY

PROBLEM STATEMENT: Create a NumPy array of message lengths and reshape it into a matrix.



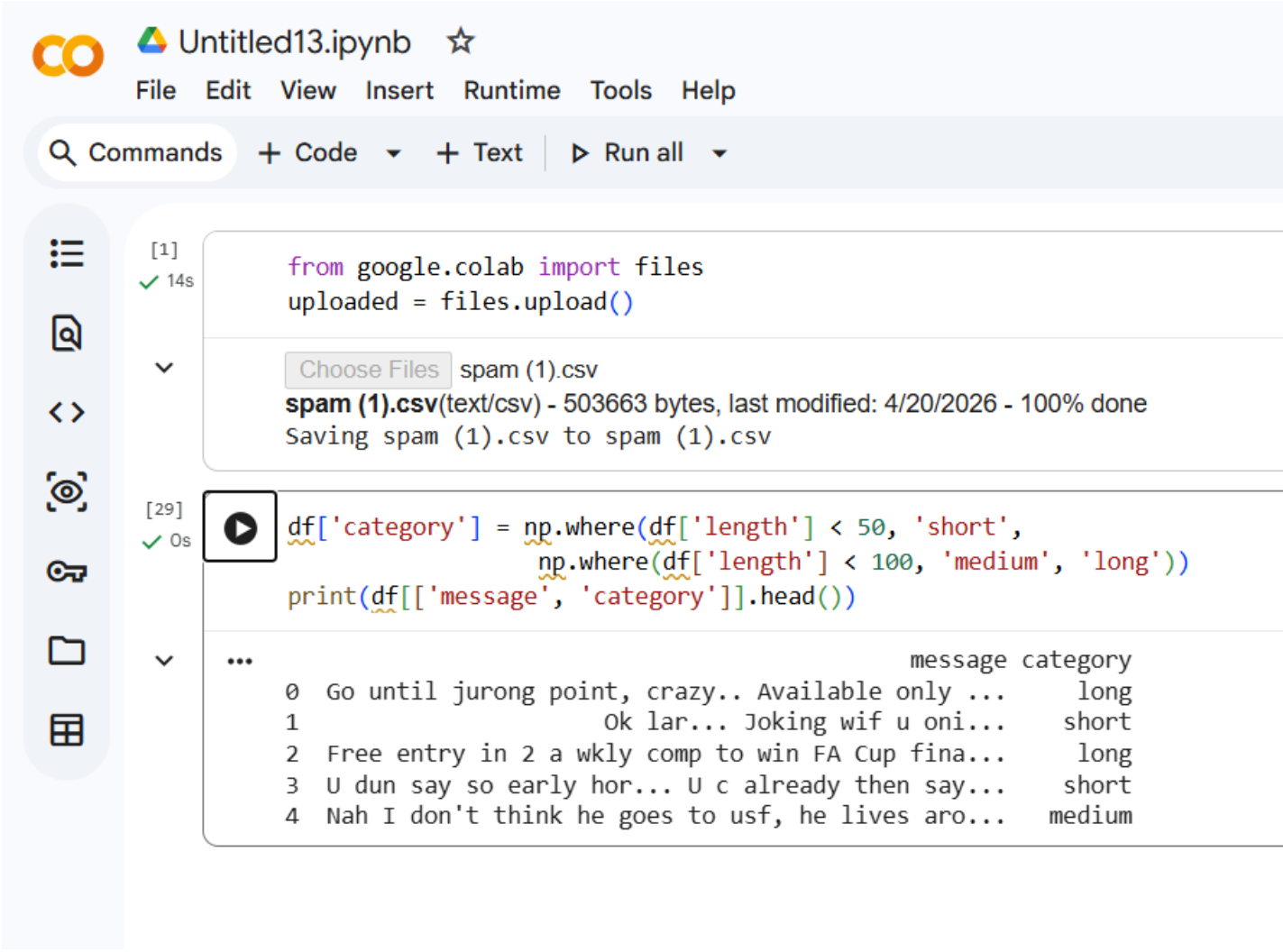
13. MESSAGES > 100 CHARACTERS

PROBLEM STATEMENT: Perform element-wise comparison to identify messages longer than a threshold (e.g., >100 characters).



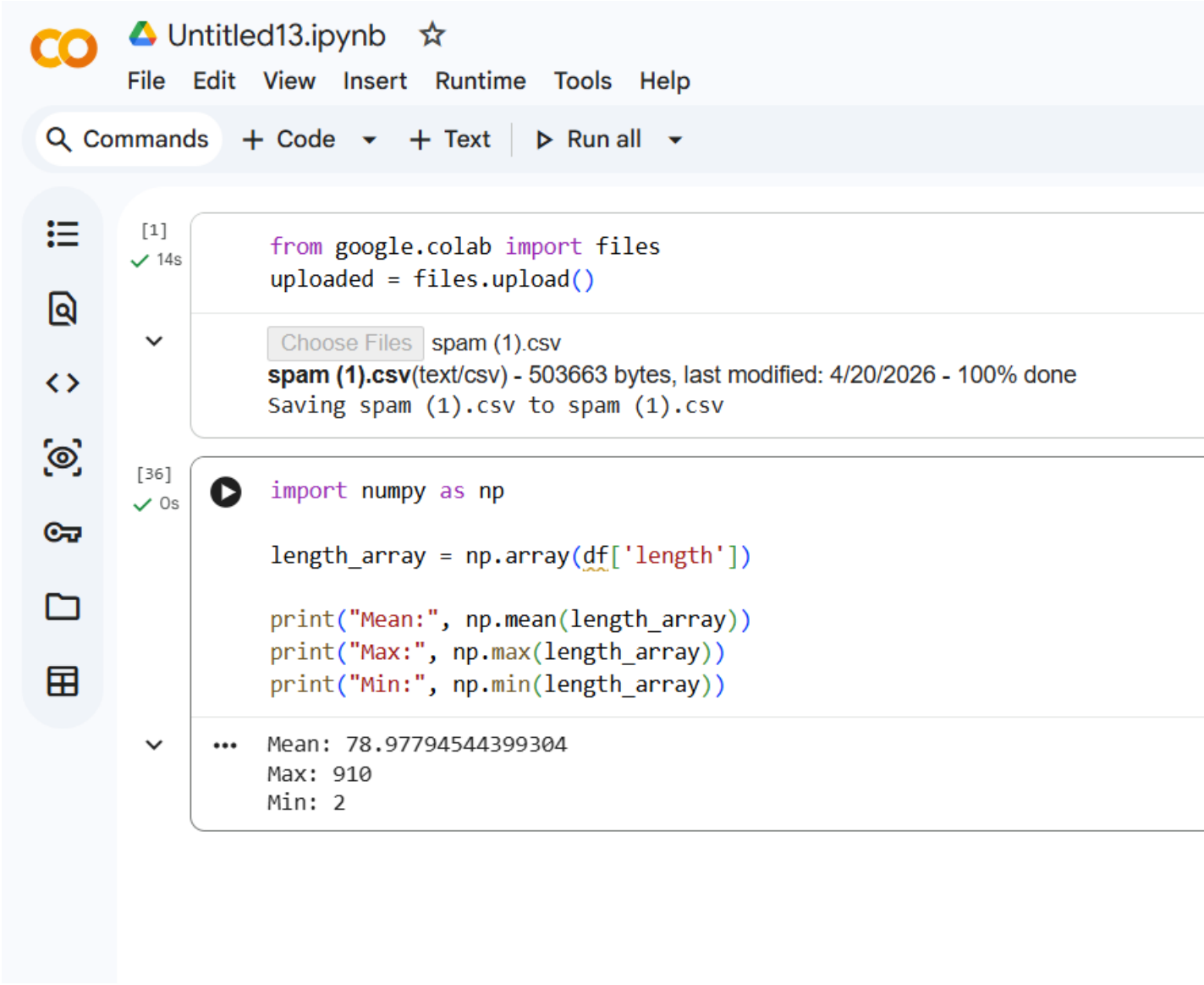
14. CATEGORIES MESSAGE LENGTH

PROBLEM STATEMENT: Use NumPy where conditions to classify messages as “short”, “medium”, or “long”.



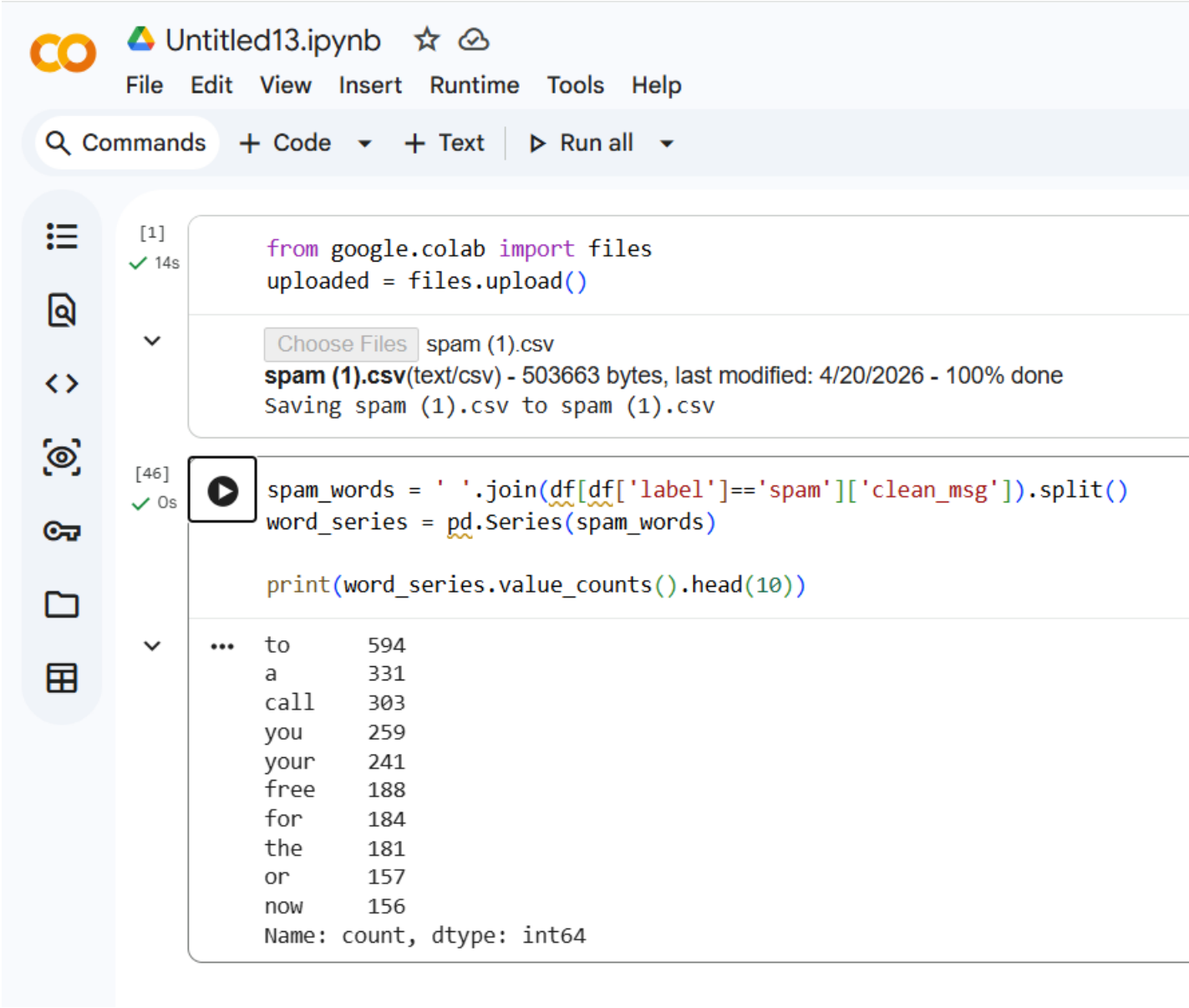
15. NUMPY OPERATIONS

PROBLEM STATEMENT: use the NUMPY library to perform basic statistical calculations on dataset



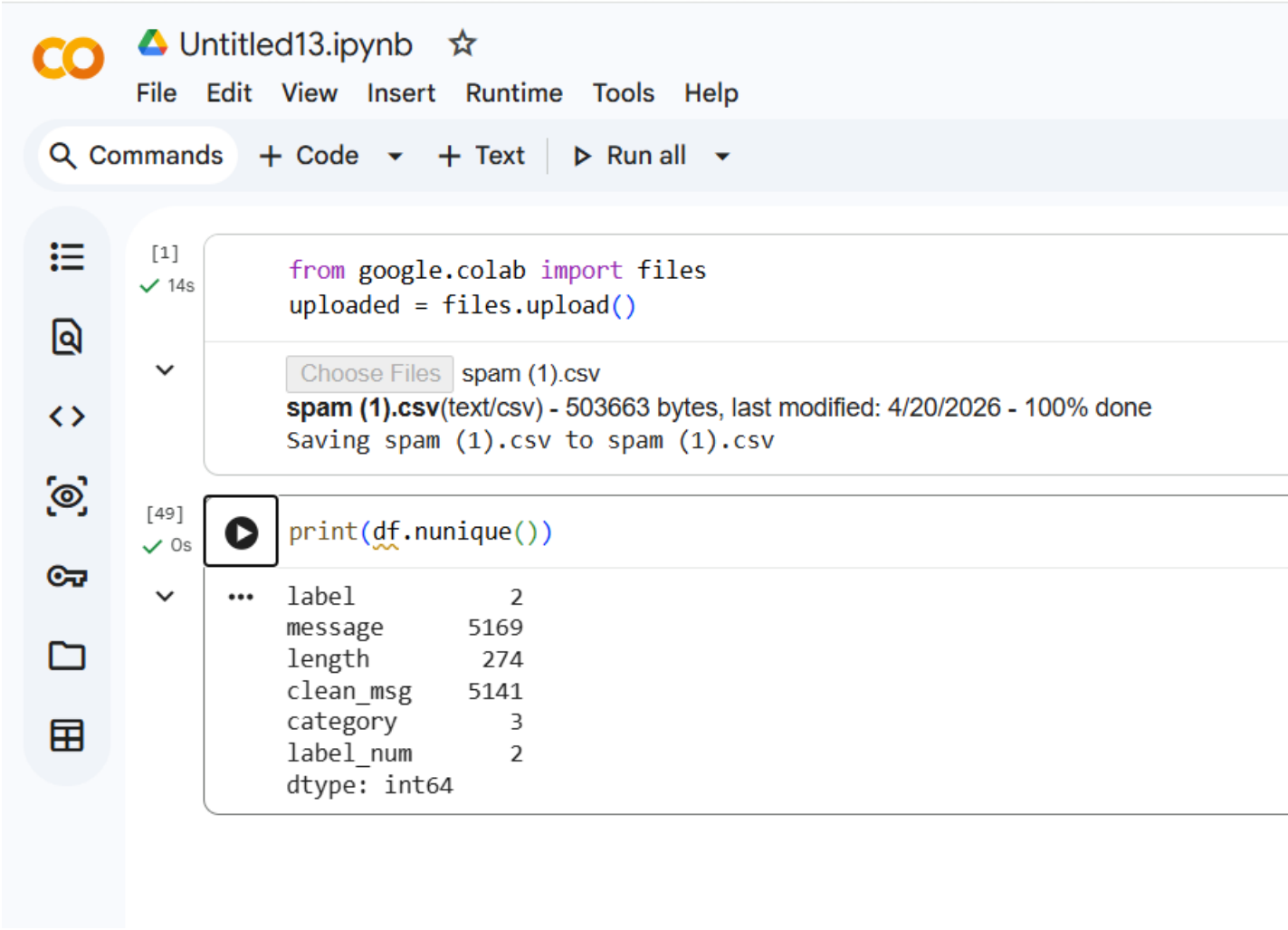
16. WORD FREQUENCY ANALYSIS

PROBLEM STATEMENT: extract all words from messages labelled as “spam”.



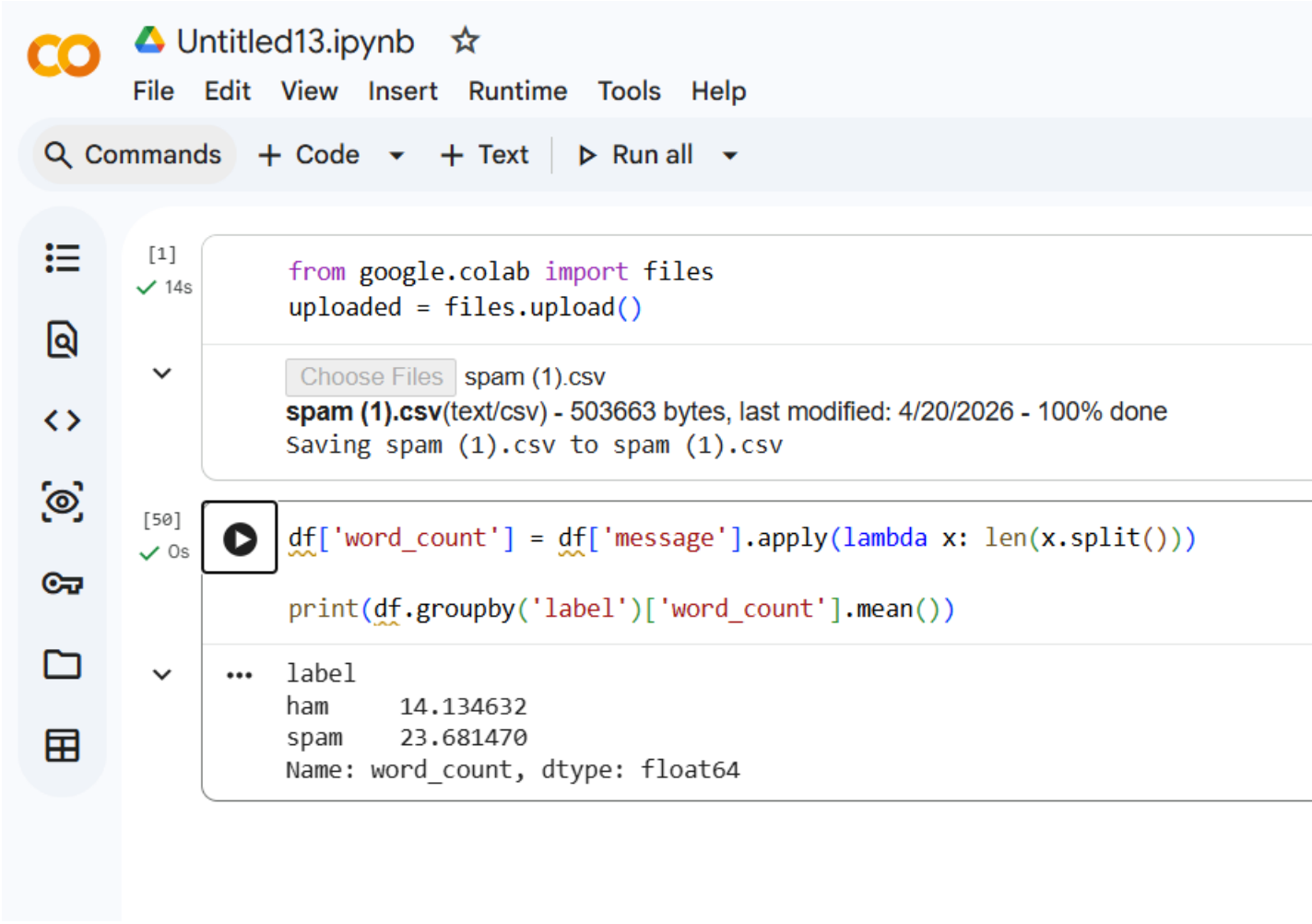
17. UNIQUE VALUES

PROBLEM STATEMENT: Count how many messages are labeled as “spam” and “ham



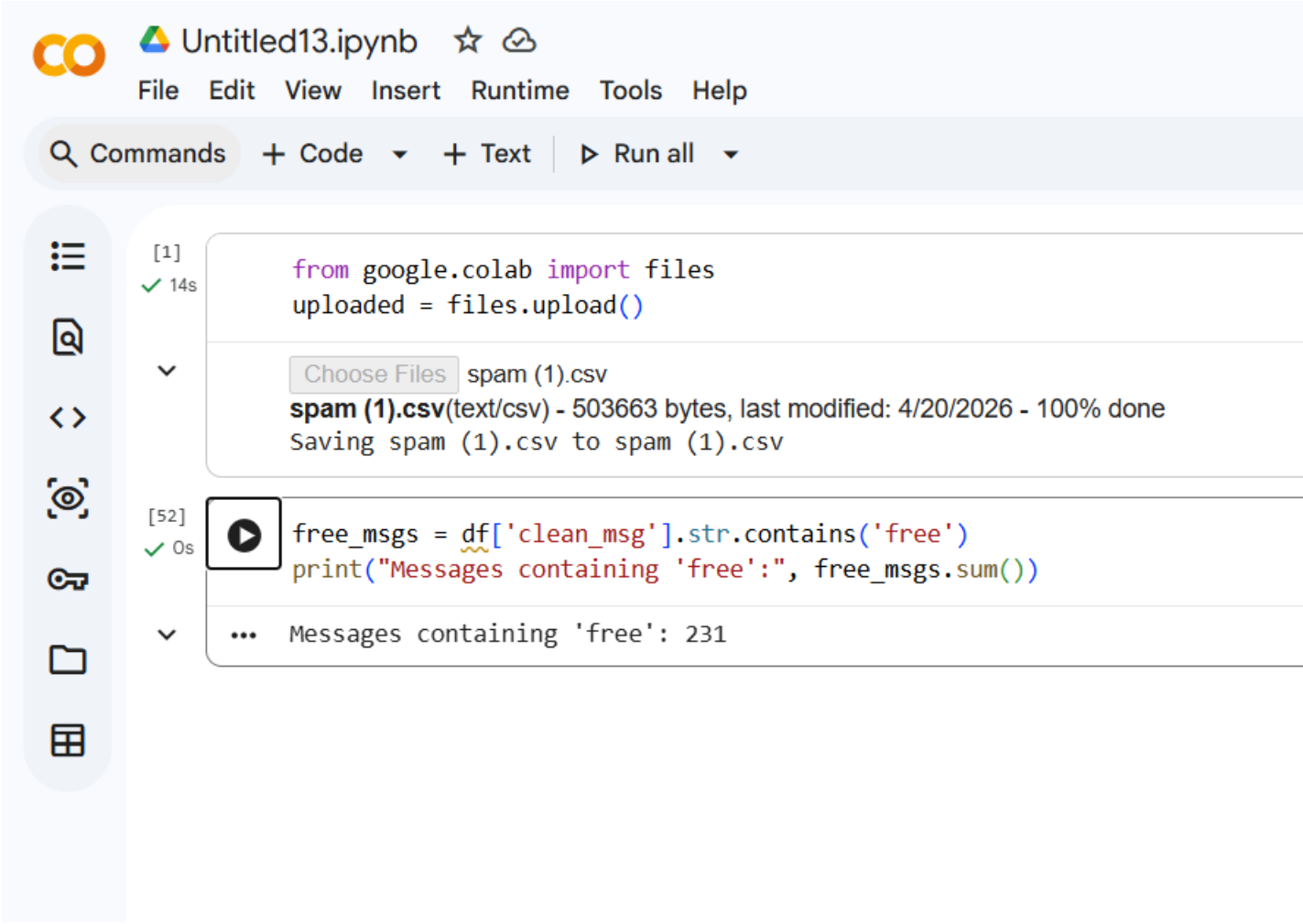
18. Find Average Word Count in Spam vs Ham Messages

Problem Statement : Calculate average number of words in spam and ham messages



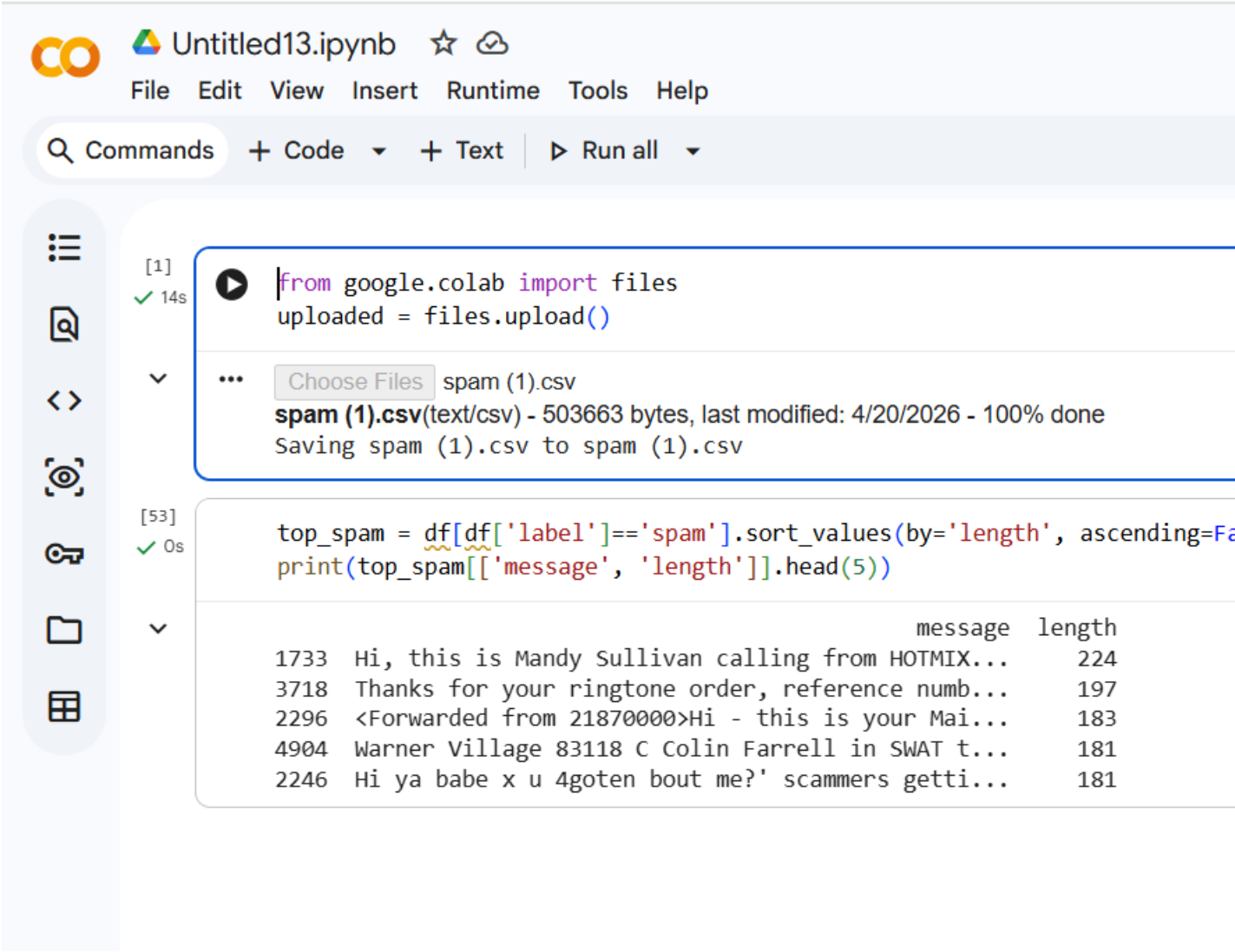
19. Find Messages Containing Specific Keyword ("free")

PROBLEM STATEMENT: Count how many messages contain the word “free”



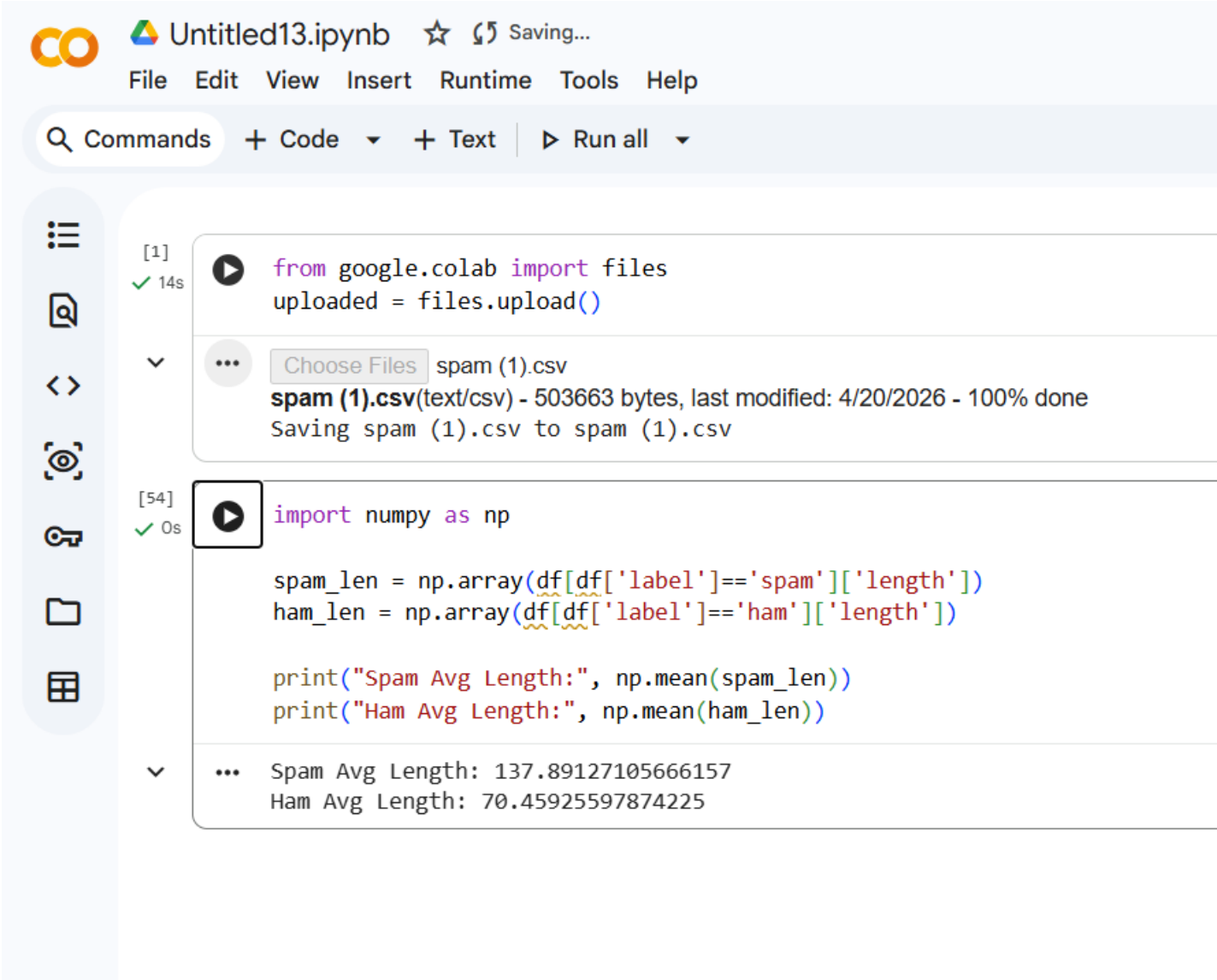
20. Find Top 5 Longest Spam Messages

PROBLEM STATEMENT: Display top 5 longest spam messages



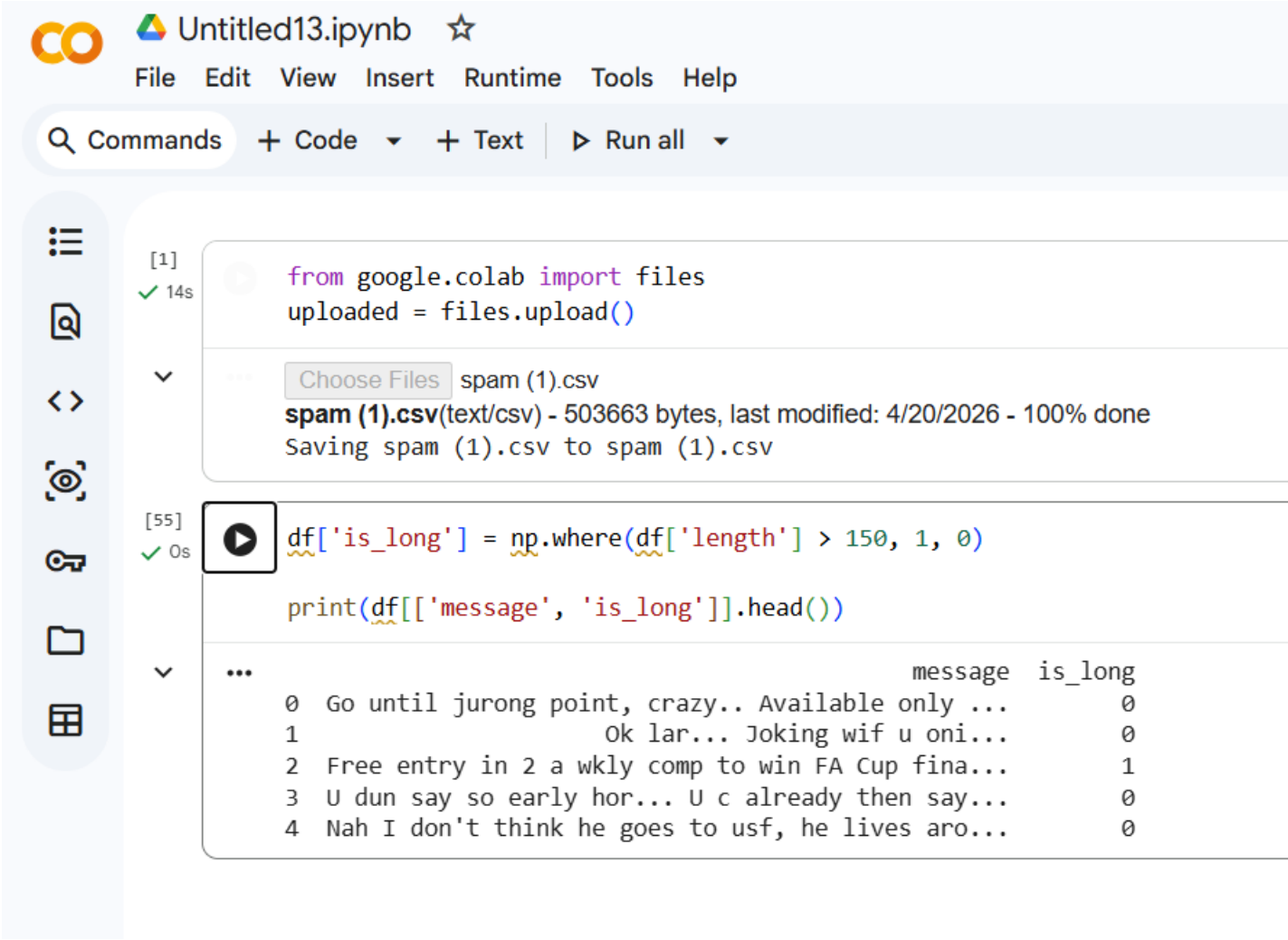
21. Compare Average Length Using NumPy (Spam vs Ham)

PROBLEM STATEMENT: Use NumPy to compare average lengths



22. Create Binary Feature for Long Messages (>150 chars)

PROBLEM STATEMENT: Classify messages as long (1) or short (0)



Result / Observation :

- Dataset was successfully loaded and analyzed using NumPy and Pandas.
- Class distribution shows 86.6% ham and 13.4% spam — clear class imbalance.
- Spam messages are on average nearly 2x longer than ham messages (138 vs 71 chars).
- Spam messages contain more digits, uppercase letters, punctuation, and URLs.
- NumPy boolean masking and z-score methods efficiently identified outlier messages.
- Feature engineering columns (length, word count, digit count) can be used for ML.

Conclusion :

- NumPy and Pandas provide efficient and expressive methods for data analysis.
- The SMS Spam dataset clearly shows distinct statistical patterns between spam and ham.
- Feature engineering with Pandas str methods and NumPy math operations is powerful.

- Data cleaning (removing duplicates, null checks) is essential before analysis.
- Message length, digit count, and uppercase ratio are strong indicators of spam.